FlightFlow: A Vision-Based Smart Airport Queue and Resource Optimization System

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***Abstract*—Long queues at airport check-in counters and security lanes delay passengers and make staff planning difficult. This paper introduces FlightFlow, a Python prototype that combines computer vision, agent-based simulation, and a web dashboard to measure queues and to test changes before they are made. People in a video are detected with YOLO and followed with the ByteTrack tracker, and the tracks are converted into queue length, waiting time, counter occupancy, and entry and exit counts. A simulation of passengers moving through check-in, security, optional immigration, and the departure gate is implemented in both NetLogo and Python. Users can change the number of counters, lanes, or the arrival rate and compare the outcome with a baseline using identical random seeds. The paper gives the motivation, a short review of related work, and the design of the system. The review shows that waiting-time measurement, people tracking, and passenger-flow simulation are usually studied separately. In a simulated example with assumed demand, two extra check-in counters reduced the mean wait from 1.43 to 0.37 minutes, while one extra security lane made little difference. These numbers are simulated and are not measurements from a real airport. The accuracy of the camera module on real footage still has to be validated.**

***Index Terms*—airport queue management, computer vision, YOLO, ByteTrack, agent-based modeling, decision support.**

I. INTRODUCTION

Every passenger must pass several steps at an airport: check-in, security, immigration, and boarding. If one step becomes slow, a queue forms and the delay moves to the next step. Long queues can make passengers miss their flights, and staff then have to be moved quickly from one place to another.

Most airports already have cameras, but a camera only shows how many people are standing in a line at one moment. It does not tell how long each person has waited, and it cannot show what would happen if a new counter were opened. A simulation can answer that question, but it depends on assumed numbers. So we decided to use both methods together.

For the camera part, YOLO finds people in a frame in a single pass [1], and ByteTrack keeps the same ID for a person across frames, even when the person is partly hidden [2]. For the simulation part, agent-based modeling treats each passenger and each desk as a separate agent, and queues form from their interaction.

This paper presents FlightFlow, which connects these ideas in one system. Its main contributions are: (i) a tracking module that measures waiting time and counts people entering and leaving, (ii) an airport simulation in NetLogo and Python, (iii) a tool for comparing scenarios, and (iv) a dashboard that clearly separates observed, simulated, and demo data. FlightFlow is a prototype for decision support and is not a certified airport system. Section II reviews related work.

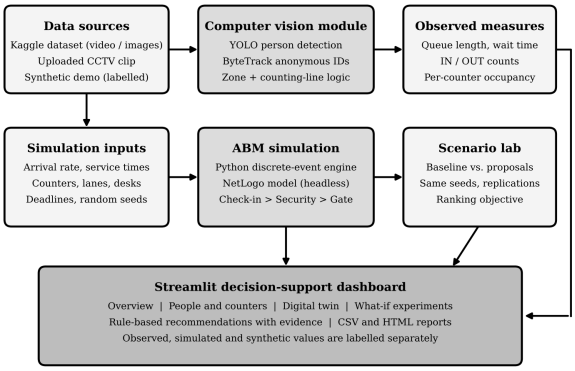


Fig. 1. Architecture of FlightFlow: data sources feed the computer vision module and the agent-based simulation, and both report to one decision-support dashboard.

II. LITERATURE REVIEW

A. Measuring Waiting Times at Airports

Waiting time at security has traditionally been measured by hand, for example by giving passengers time-stamped cards at the start of the queue and collecting them at the end. Bullock et al. replaced this with an anonymous Bluetooth system at Indianapolis International Airport that recorded when a device passed readers at the entrance and exit of the screening area [3]. Their work showed that automatic measurement is practical and useful for staffing decisions. However, it only follows passengers who carry a detectable device, and it gives a travel time rather than a picture of the queue itself, such as how many people are waiting or which counter is busy. Video can provide this missing detail.

B. People Detection and Tracking

Modern detectors such as YOLO are fast enough to process video frame by frame [1]. Tracking-by-detection then links the boxes over time so that each person keeps one identity. ByteTrack improves this step by keeping low-confidence detections for a second matching round, which recovers people who are partly hidden behind others [2]. This is common in queues, so it suits our problem. Methods that only count people per image are useful in very dense scenes, but they cannot give an individual waiting time, which is why FlightFlow follows each person with a

tracker.

C. Simulation of Passenger Flow

Agent-based models have been used to study how passengers move through airport processes. Cheng et al. built an agent-based model of the international departure process, from check-in and security to customs and boarding, and used it to study how groups of travellers affect the check-in process [4]. As with any simulation, such results depend on the input values chosen, so they are most useful when those inputs come from real observation.

D. Research Gap

Table I summarises the approaches. Waiting-time sensors, video tracking, and simulation each solve part of the problem. In the studies we reviewed, they are reported separately, and we did not find a small open tool that carries one queue from video measurement to a ranked what-if comparison. Our review is short and not exhaustive, but this gap motivated FlightFlow.

TABLE I. SUMMARY OF RELATED APPROACHES

Approach	**Strength**	**Limitation**
Bluetooth timing [3]	Automatic travel time	Only device carriers; no queue detail
Detection + tracking [1], [2]	Per-person position and time	Occlusion, ID switches
Agent-based simulation [4]	Tests what-if changes	Depends on assumed inputs
FlightFlow	**Links video and simulation**	**Not yet validated on real data**

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